

Boyang Zhou

Ph.D. Student, Wireless@Virginia Tech
boyangzhou@vt.edu — (814) 954-9288 — linkedin.com/in/byzhou1 — boyang-zhou.github.io

Summary

Fifth-year Computer Engineering Ph.D. student at Wireless@Virginia Tech. Specializing in wireless communications across the PHY/MAC domains. Applies communication theory, mathematical optimization, and machine learning to the design and optimization of reliable, resource-efficient wireless systems. Expertise spans cellular PHY/MAC control procedures, link-level performance analysis, digital signal processing (DSP) algorithm design, and MAC-layer scheduling.

Education

Ph.D. in Computer Engineering, Virginia Tech Advisor: Y. Thomas Hou	Aug. 2022 — Present
M.S. in Computer Engineering, Virginia Tech GPA: 3.9/4.0	Aug. 2022 — Dec. 2024
B.S. , The Pennsylvania State University Dual Degree in Computer Science and Mathematics GPA: 3.91/4.00, <i>Magna Cum Laude</i>	Aug. 2018 — May 2022

Work Experience

Graduate Research Assistant <i>Wireless@Virginia Tech</i>	Aug. 2022 — Present
- Designed PHY/MAC control and radio-resource management algorithms for multiuser wireless systems, accounting for practical cellular control procedures and PHY conditions including imperfect CSI and multiuser interference, while satisfying stringent transmission-reliability requirements. - Developed a 3GPP-based multiuser MIMO-OFDM link-level simulation incorporating channel-estimation uncertainty, transmit beamforming/precoding, modulation, channel coding/decoding, and link-performance evaluation. - Designed real-time ML-assisted beamforming and MAC scheduling algorithms under practical PHY/MAC constraints, including compact neural-network models for SINR ranking and transmission-feasibility classification. - Utilized communication theory, real-time GPU algorithm design, convex optimization, machine learning, and reinforcement learning to address PHY/MAC layer resource allocation problems under sub-5-ms execution-time constraints. - Performed RF-aware wireless system analysis by accounting for RF impairments and transceiver nonidealities and characterizing their effects on baseband signal processing, multiuser interference, effective SINR, and spectrum utilization.	

Projects

MIMO-OFDM Link-Level Simulator

- Developed a MATLAB-based link-level simulator for a MIMO-OFDM system.
- Modeled AWGN and time- and frequency-selective multipath channels with Rayleigh, Rician, and Nakagami- m fading, including delay spread, Doppler spread, and spatial antenna correlation.
- Implemented OFDM modulation and demodulation, cyclic-prefix insertion and removal, pilot-based LS channel estimation, channel equalization, error-correction coding, interleaving, and adaptive subcarrier bit loading.
- Implemented OFDM carrier-frequency and symbol-timing synchronization to estimate and compensate for carrier-frequency and timing offsets.
- Implemented MIMO transmission and reception techniques, including MRT transmit beamforming, MRC receive combining, SVD-based spatial multiplexing, Alamouti space-time block coding, and ZF, MMSE, SIC, and maximum likelihood detection.
- Evaluated BER, BLER, throughput, and spectral efficiency and quantified the trade-off between pilot overhead and transmission performance.

Radio Resource Management Using Parallel and Concurrent Programming

- Designed and implemented a parallel MAC-layer resource-allocation algorithm in C/C++, using OpenMP multithreading, MPI message passing, and CUDA GPU programming, including shared-memory flags and block-level synchronization for thread coordination.
- Explored CPU memory locality through loop reordering, matrix transposition, and cache tiling, and coordinated GPU threads using shared-memory flags and block synchronization.

Reinforcement Learning

- Implemented and evaluated TD(λ), Q-learning, projected policy gradient, natural policy gradient, actor-critic, Proximal Policy Optimization (PPO), and PPO- λ in Python/PyTorch.

- Implemented tabular value/Q-function learning, Fourier-basis linear function approximation, and neural-network-based actor-critic policy/value approximation, and evaluated them on standard OpenAI Gym environments including Cliff Walking, CartPole, and Continuous Mountain Car.
- Tuned learning rate, discount factor, and related hyperparameters using Optuna and compared algorithms based on convergence behavior, episodic reward, and training stability.

End-to-End RAN Prototype Using MIMO (Team Project)

- Designed and implemented a QoS-aware MAC-layer scheduler for an end-to-end 2×2 MIMO RAN prototype for next-generation cellular networks.
- Designed the scheduler to allocate radio resources based on application throughput and latency requirements and real-time feedback from the RAN.
- Evaluated the scheduler in terms of throughput, latency, and packet loss on the end-to-end prototype.

End-to-End Software-Defined Cellular Network Deployment and Control

- Deployed an end-to-end software-defined cellular system using the NSF Open AI Cellular (OAIC) platform and verified connectivity across the core network, base station, user equipment, and virtual radio link.
- Configured an Open Radio Access Network (O-RAN) near-real-time RAN Intelligent Controller (near-RT RIC) and connected it to an **srsRAN** base station for network monitoring and control.
- Conducted hands-on experiments with near-real-time RAN control applications (xApps) for key performance indicator (KPI) monitoring, radio-resource scheduling, and AI-driven network control using OAIC's controller and automated testing frameworks.

Publications

Conference Papers

- **B. Zhou** (First author) [redacted] Coauthors and title redacted for double-blind review
Submitted to *IEEE INFOCOM 2027*
- **B. Zhou** (Second author) [redacted] Coauthors and title redacted for double-blind review
Submitted to *IEEE INFOCOM 2027*

Journal Papers

- **B. Zhou**, Y. Shi, Y. Wu, S. Acharya, L. DaSilva, W. Lou, and Y. T. Hou, "Atalanta: A Bandwidth-Conserving URLLC Scheduler for Industrial Automation," under minor revision for *IEEE Internet of Things Journal*.
- Q. Liu, C. Li, S. Li, Y. Shi, **B. Zhou**, Y. T. Hou, and W. Lou, "Pistis: A Scheduler to Achieve Ultra Reliability for URLLC Traffic in 5G O-RAN," *IEEE Internet of Things Journal*, vol. 11, no. 21, pp. 35405–35419, November 2024.
- J. Zhao, **B. Zhou**, J. P. Butler, R. G. Bock, J. P. Portelli, and S. G. Bilén, "IoT-based sanitizer station network: A facilities management case study on monitoring hand sanitizer dispenser usage," *Smart Cities*, vol. 4, no. 3, pp. 979–994, July 2021.

Video Demo

- CNSR@VT, "A Demo of Data Scheduling for Ultra-Reliable Low Latency Communications in O-RAN," *Video*, YouTube, 2024. [Online]. Available: <https://www.youtube.com/watch?v=LNp8X76iEpA>.

Selected Honors and Awards

First Place, *Open6G+AI Workshop Hackathon*

Apr. 2026

- Won first place in a team competition with participants from more than 30 institutions worldwide.
- Helped design the MAC-layer scheduler around application QoS requirements and RAN feedback; contributed to its implementation and integration; and diagnosed deployment and connectivity failures during end-to-end testing.
- Competition organized by Northeastern University's Institute for Intelligent Networked Systems.

Cyber Innovation Scholar, *Commonwealth Cyber Initiative Southwest Virginia*

Apr. 2026

- Selected to explore the practical impact and commercialization potential of cyber and wireless research.
- Received \$1,500 in professional-development funding for conference travel and technical training.
- Completed research-commercialization and entrepreneurship training through a cyber startup workshop led by Virginia Tech LAUNCH.

Department Research Fellowship, *Virginia Tech ECE*

Aug. 2022

- Selected as the sole fellowship recipient among over 50 incoming Ph.D. students in the ECE Department.
- Received approximately \$70,000 in funding covering full tuition and a stipend for the first year of the Ph.D. program.

First Place, *CodePSU Programming Competition, Beginner Division*

Mar. 2019

- Won the beginner division of a campus-wide algorithmic programming competition.
- Competition organized by Penn State's student chapter of the Association for Computing Machinery.

Other Experiences

Math Guided Study Group Leader

Mar. 2020 — May 2022

The Pennsylvania State University

- Reviewed the material students learned during the past week.
- Communicated with professors and students to go through students' questions.

Math Tutor

Sep. 2019 — May 2022

The Pennsylvania State University

- Weekly assisted more than 15 students one-on-one in courses ranging from trigonometry to Calculus II.

Skills

Programming:	C, C++, CUDA, Python, Pytorch, MATLAB, Verilog, Java, SQL, Git
Software:	Adobe Photoshop, After Effects, Premiere
Languages:	English, Chinese, French (B1 level)